

9.4.1 yum: Yellowdog Updater, Modifier

yum is used to update the rpm-based packages, meaning Red Hat / Fedora, SUSE and Mandriva, besides other distros.

How yum Works:

An rpm package consists of what is called a header along with the archive of the program itself. The header contains the list of files the make up the archive, a description of the package with the libraries it provides and requires from other packages and if it has any known conflicts with other packages. The package manager requires this information so that when a package is installed or uninstalled, none of the existing packages are affected. This process is called “resolving package dependencies”. It is a challenging task for any package management tool. A situation may arise such that a packaged tool requires two or three libraries (a library is a list of commands or functions that are often used). These libraries in turn may require some other libraries. By the time the package is installed, there may be a lot more packages, and none of them must be conflicting with the other.

yum maintains a list of URL from which the repository is accessed in order to upgrade the packages when called for. This list is usually kept in a file called `yum.conf` located in the `/etc` directory. It is possible to edit this file to make changes to the way yum behaves while updating. You can also add URLs other than what is included by the distributions. Let us look at basic yum commands and then some of the changes you can make to `yum.conf`

To get yum to install a package the command is as follows:

yum install <package>

(replace “*package*” with the actual name of the package).

Yum checks if *package* is already installed and is if, the version. If it finds the package not present, it downloads the package and installs it.